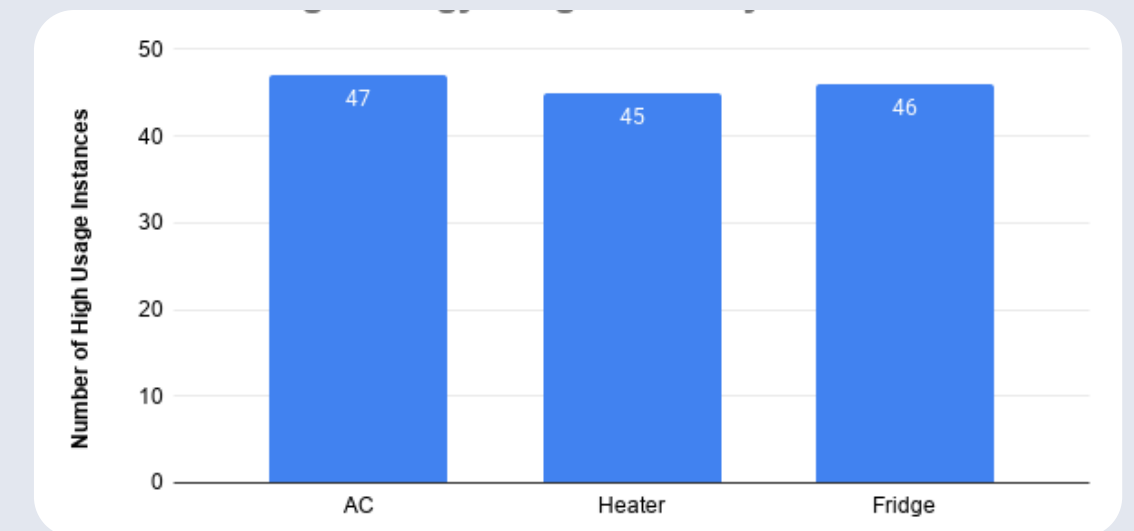
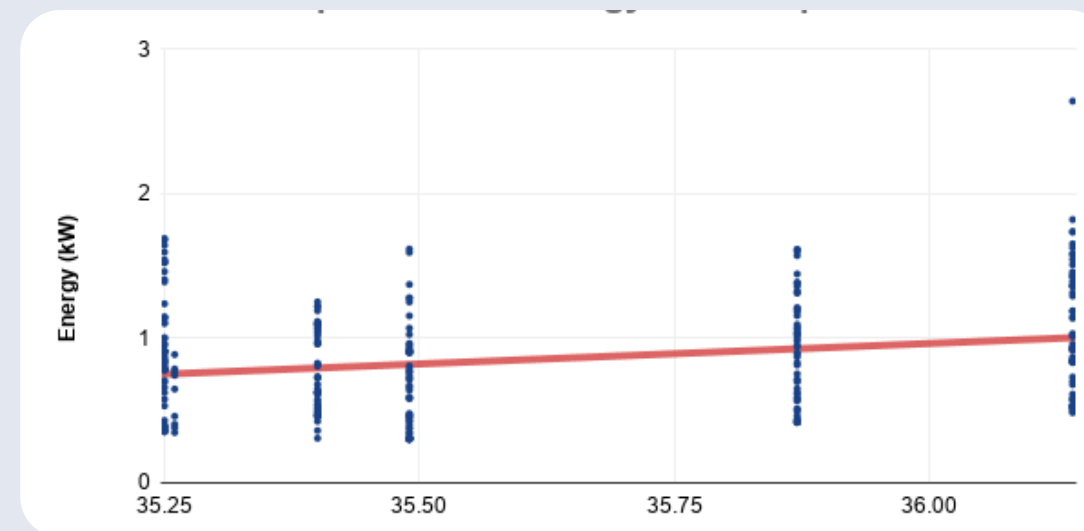
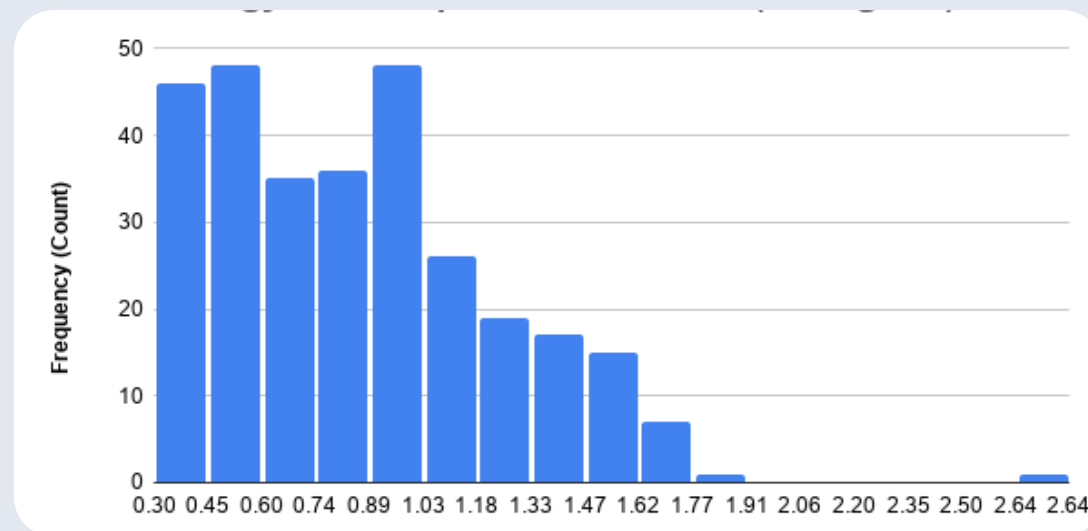


Smart Home Energy Consumption Analysis

Data Analysis using Excel



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Introduction



- Smart homes generate continuous energy usage data
- Understanding consumption patterns is important for efficiency
- Data analysis helps identify trends and anomalies
- This project analyzes energy consumption using Excel tools



Dataset & Methodology



Dataset

- Source: Kaggle Smart Home Dataset
- Parameters:
 - Time (timestamp)
 - Energy (kW)
 - Temperature (°C)
 - Humidity
- Time-series data collected from smart home sensors

Methodology

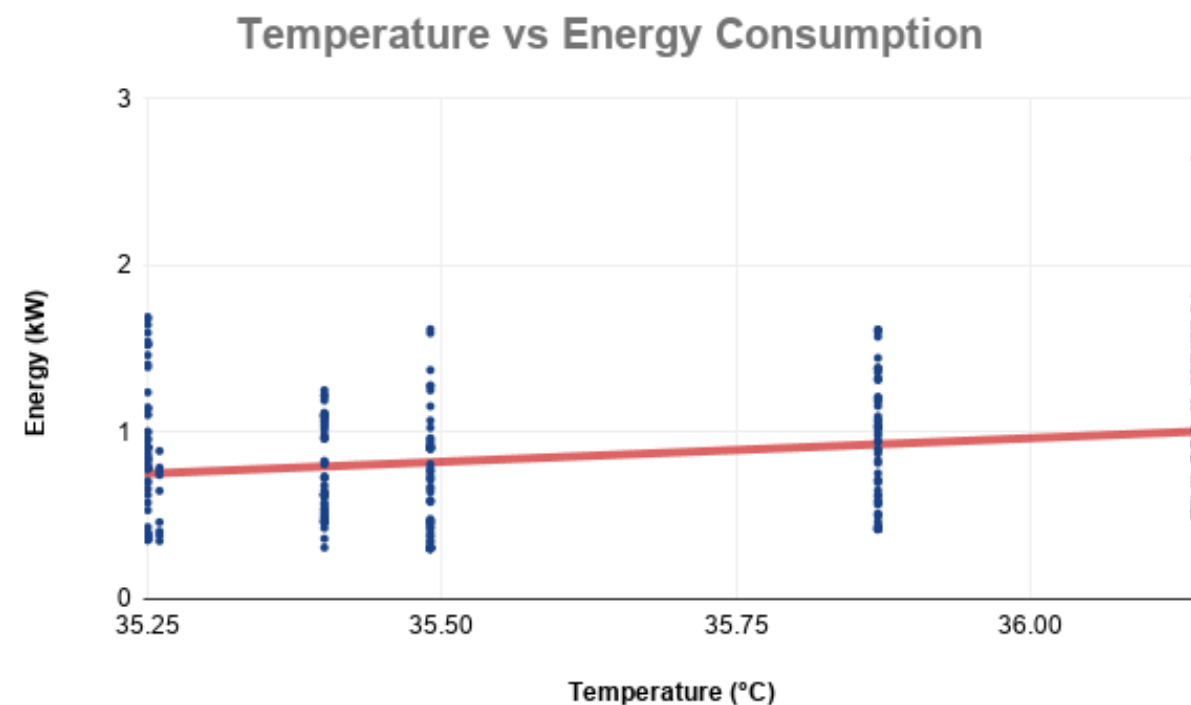
- Data cleaning and preprocessing
- Feature extraction (Hour, Time Blocks)
- Energy scaling for normalization
- Classification using threshold (IF logic)
- Device mapping using VLOOKUP
- Time segmentation using HLOOKUP

Data Analysis & Results

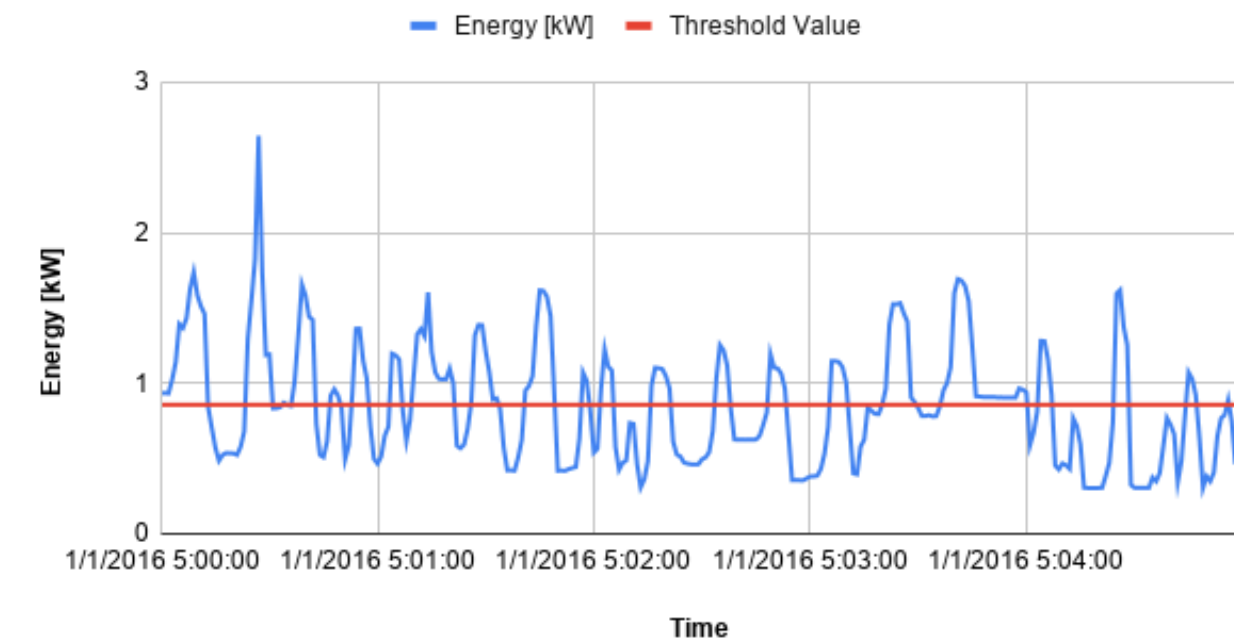


Temperature vs Energy scatter plot

- Shows relationship between temperature and energy
- Weak correlation observed
- Energy consumption not strongly dependent on temperature



Energy Consumption Over Time with Threshold



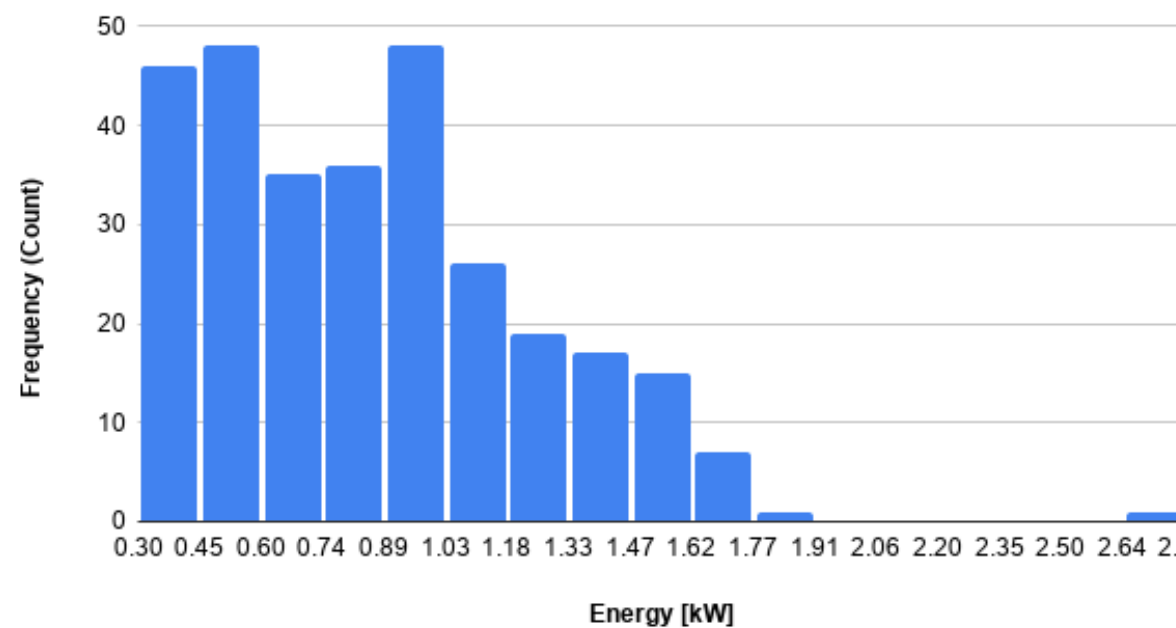
Energy vs Time graph

- Shows variation of energy over time
- Threshold line indicates high usage
- Helps identify peaks in consumption

Distribution & Device Analysis



Energy Consumption Distribution (Histogram)



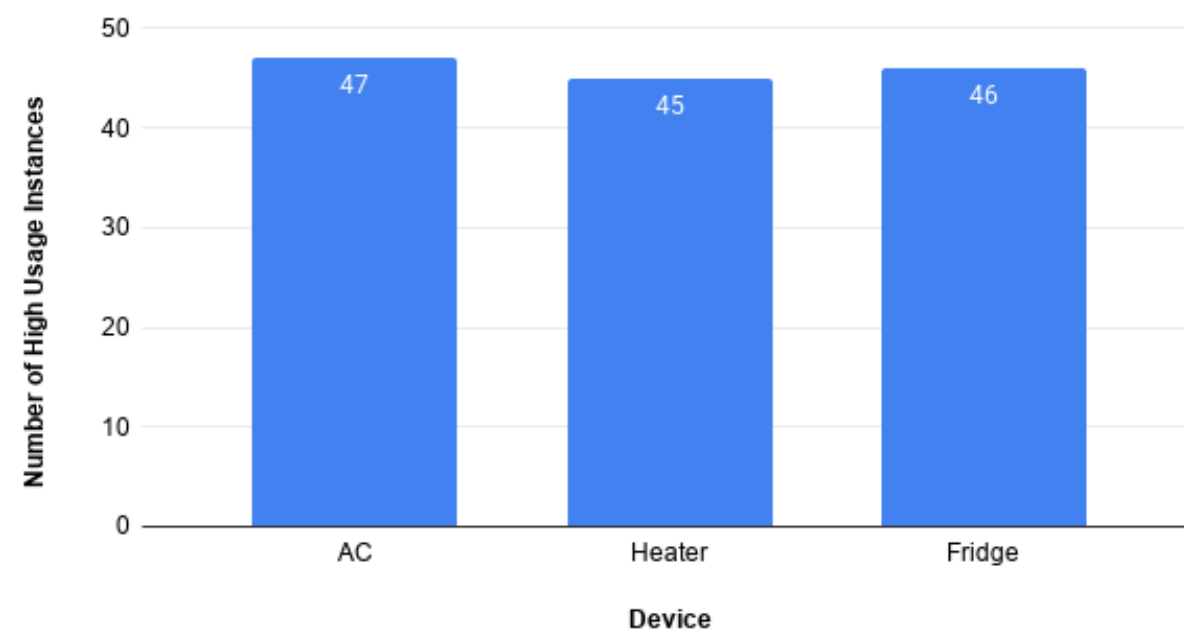
Energy Histogram

- Shows distribution of energy consumption
- Most values lie in moderate range
- Indicates stable usage pattern

Usage Level Distribution



High Energy Usage Count by Device



Device-wise High Usage bar chart

- Compares high usage across devices
- Shows variation between AC, Heater, and Fridge
- Helps identify energy-intensive devices

High vs Normal usage pie chart

- Shows proportion of High vs Normal usage
- Majority of usage falls in Normal category
- Helps understand consumption behavior

Conclusion



- Energy consumption shows clear variation over time
- Weak correlation between temperature and energy usage
- Majority of usage falls under Normal category
- Device-wise analysis highlights usage differences
- Threshold-based classification effectively categorizes consumption

This analysis demonstrates how simple Excel tools can extract meaningful insights from real-world data.

Thank You